Elmo: Recursive Virtual Payment Channels for Bitcoin

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***Abstract*—A dominant approach to solving the scalability problem in blockchain systems has been the development of layer 2 protocols and in particular payment channel networks (PCNs) such as the Bitcoin Lightning Network (LN). Routing payments over LN requires the coordination of all path intermediaries in a multi-hop round trip that encumbers the layer 2 solution both in terms of responsiveness as well as privacy. The issue is resolved by *virtual channel* protocols that, capitalizing on a suitable off-chain setup operation, enable the two endpoints to engage as if they had a direct payment channel between them. Once the channel is not needed anymore, it can be optimistically closed in an off-chain fashion.**

Apart from communication efficiency, virtual channel constructions have three natural desiderata. A virtual channel constructor is *recursive* if it can also be applied on pre-existing virtual channels, *variadic* if it can be applied on any number of pre-existing channels, and *symmetric* if it imposes equitable costs on all channel participants in every execution scenario. We introduce Elmo, the first Bitcoin-suitable recursive variadic virtual channel construction. Furthermore our virtual channel constructor is symmetric and offers optimal round complexity for payments, optimistic closing and unilateral closing. We express and prove the security of our construction in the universal composition setting, using a novel induction-based proof technique of independent interest. As an additional contribution, we implement a flexible simulation framework for on- and off-chain payments and compare the efficiency of Elmo with previous virtual channel constructors.

***Index Terms*—Payment channels; Blockchain; Security**

1. Introduction

The popularity of blockchain systems has exposed their inherent scalability limitations. In particular, Bitcoin transactions incur high latency (approximately 1h [20]) and low throughput (on average 7 transactions per second [8]), properties that are inherent [8] and fundamentally constrain their suitability for frequent or time-sensitive payments.

To address these limitations, pairs of parties (the *endpoints*) can lock funds with a single on-chain transaction, thus establishing a *payment channel*, and subsequently exchange any number of payments *off-chain*, i.e., by simply exchanging direct messages and without adding any transactions to the underlying blockchain. The endpoints use the blockchain only to resolve disputes over their payment channel. When no longer needed, closing a payment channel involves posting $O(1)$ transactions on-chain. Typically, if the endpoints collaborate closing is cheaper, but either party can unilaterally close in case its *counterparty* misbehaves.

Multiple overlapping payment channels can be combined to form a payment channel network (PCN). See, e.g., [15] for a survey of PCNs. The prime PCN on Bitcoin is the Lightning Network (LN) [23].

A fundamental limitation of existing PCNs is that off-chain payments between non-adjacent parties (i.e., parties that do not share a payment channel, a.k.a. multi-hop payments) require the active participation of all *intermediaries* along a path of channels. This design introduces latency, creates availability dependencies, and leaks the size and timing of every payment.

Virtual payment channels have been proposed as a remedy to this problem. By performing an one-time off-chain setup over a sequence of preexisting channels, the two endpoints establish a virtual channel which can then be used for direct payments between them with complexity – in the optimistic case – independent of the path length. Once the virtual channel is no longer needed, it can be closed off-chain if the involved parties cooperate, or unilaterally on-chain in case of disputes. Intermediaries only need to interact during setup and cooperative closing, both of which can be done fully off-chain. Payments only involve messages between the two endpoints, just like those over adjacent parties in a PCN (cf. Table 1).

	PCN	Virtual channel
Open/Close	On-chain	Off-chain
Payment	Needs Intermediaries	No Intermediaries

Table 1: PCNs vs Virtual Payment Channels

While conceptually appealing, constructing virtual channels over Bitcoin is challenging. Bitcoin does not support Turing-complete smart contracts or covenants [22] and thus severely restricts the design space of off-chain protocols. Existing constructions either rely on expressive smart contract platforms [11], [12], impose fixed channel lifetimes [16], require excessive collateral [2], cannot be stacked in a

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recursive manner [2], or can only combine two underlying channels (i.e., is *binary*) [1]. See Sec. 5 for related work.

A virtual channel constructor can be thought of as an *operator* over a collection of payment channels that produces a new virtual channel. The preexisting channels are called *base* channels. We identify three natural desiderata for the virtual channel.

- *Recursive*. A recursive virtual channel constructor can operate over base channels that themselves could be the result of previous applications of the operator. This allows building new virtual channels on top of preexisting virtual channels, allowing the channel structure to evolve dynamically.
- *Variadic*. A variadic virtual channel constructor can be built on top of any number of base payment channels directly. This allows building a virtual channel between endpoints with multiple intermediaries with a single setup, contrary to a *binary* constructor.
- *Symmetric*. A symmetric virtual channel constructor offers setup and closing operations that are symmetric in terms of computation, network and storage cost between the two endpoints or the intermediaries (but not necessarily a mix of both). Also, payments by the two endpoints incur the same delay. Importantly, this ensures that no party is worse-off or better-off after an application of the operator in terms of accessing the basic channel functionality, providing simple incentives.

Binary channels are also problematic as successive applications of a recursive binary virtual channel operator to connect distant endpoints breaks symmetry. This is of particular concern since most previous virtual channel constructors proposed are binary [12], [1], [16]. Therefore having variadic channels is even more important.

As already identified [12], recursiveness is an important design property, but has been achieved only for fixed lifetime channels [16] and has been left as an open question for indefinite lifetime channels on Bitcoin [1]. Recursive channels allow moving off-chain coins flexibly, quickly, with minimal interaction, and at a low cost, even when the blockchain is consistently congested. Parties can open new recursive virtual channels on an as-needed basis. Recursiveness allows the off-chain moving of coins to a new channel even if the base channel parties are unresponsive. It also enables moving only part of the channel coins to a new channel. In contrast, coins in a non-recursive virtual channel [2] must be first moved on-chain, which is slow and expensive. The flexibility of recursiveness can inspire confidence in virtual channels, prompting users to transfer more coins off-chain and reduce on-chain congestion.

A scenario that needs both the recursive and the variadic properties is as follows (Fig. 1): Initially Alice has a channel with Bob, Bob one with Charlie and Charlie one with Dave. Alice opens a virtual channel with Dave over the 3 channels – this needs the variadic property. She later realises she has to pay Eve a few times, who happens to have a channel with Dave. Alice interacts just with Dave and Eve to move half her coins from her virtual channel with Dave to a new one with Eve – this needs the recursive property. Recursive, variadic, symmetric virtual channels with indefinite lifetime over Bitcoin have remained an open problem until now.

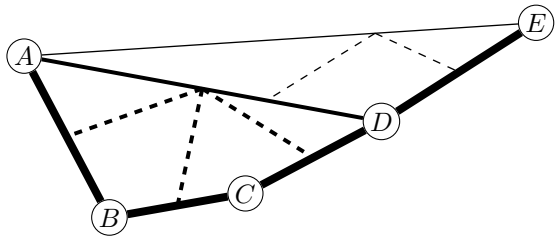


Figure 1: An Elmo network with 5 nodes, 4 *simple* (i.e., on-chain) channels (thick lines), 1 virtual channel built only on simple ones (medium line), and 1 virtual channel built on the virtual and a simple one (thin line). Each virtual channel is connected with its *base* channels with dashed lines. The variadic and recursive properties are showcased.

Our Contributions. In this work, we resolve this open problem by introducing Elmo (named after St. Elmo's fire), the first Bitcoin-compatible virtual payment channel construction that is recursive, variadic, and symmetric, while supporting channels of indefinite lifetime. We build on Bitcoin as it is the first and most established blockchain, with formal security guarantees [14].

At a high level, Elmo enables two endpoints to establish a virtual channel over an arbitrary sequence of base channels – each of which may itself be virtual – using a one-time off-chain setup protocol. Thereafter, payments proceed exactly as in a simple payment channel, with constant round complexity and without involving intermediaries. Virtual channels can be closed cooperatively entirely off-chain, or unilaterally on-chain by any participant, with at most two on-chain transactions per party and without dependence on the number of intermediaries.

A key technical challenge is avoiding the need to store an exponential number of transactions and signatures as channels are composed recursively. Elmo addresses this by leveraging the `ANYPREVOUT` signature type [9], a candidate for inclusion in the next Bitcoin update, which allows commitment transactions to be reused across different funding outputs. We conjecture that without such a mechanism, efficient recursive virtual channels over Bitcoin are impossible. For more details on `ANYPREVOUT`, see Sec. 2.

We formalise and prove the security of our protocol in the Universal Composition (UC) [5] setting (full version [17]); our *global* functionality captures both simple and virtual channels and introduces a novel induction-based argument to handle recursive composition of functionalities of independent interest. On globality, see [4] and Sec. 3. We also guarantee that the real-world protocol has *balance security* [13], i.e., that honest parties can close the channel and get the expected coins on-chain (Lemma 1). We note that privacy is not an explicit goal of Elmo.

Finally, we evaluate Elmo using a novel simulation framework for off-chain payment networks. Our results

show that Elmo achieves latency and fee performance comparable to or better than existing Bitcoin-compatible virtual channel protocols, while offering strictly greater flexibility.

In summary, we make the following contributions:

- We present Elmo, the first Bitcoin-compatible virtual payment channel construction that supports recursive and variadic composition with indefinite lifetime and symmetric costs.
- We provide a formal proof of security in the Universal Composition framework and establish real-world balance security guarantees.
- We implement a simulation framework and demonstrate that Elmo performs on par with or better than prior work while enabling strictly more expressive off-chain interactions.

2. Protocol Description

Conceptually, Elmo is split into four actions: channel opening, payments, cooperative closing and unilateral closing. Parties P_1 and P_n may **open** a channel (P_1, P_n) between them directly on-chain, in which case they follow an opening procedure like that of LN; such a channel is called *simple*. Otherwise they can open it on top of a path of preexisting *base* channels (P_1, P_2), (P_2, P_3), $\ldots$, (P_{n-1}, P_n). Then (P_1, P_n) is a *virtual* channel. A channel is either simple or virtual, not both. Elmo is recursive, thus each base channel may itself be simple or virtual. To open a virtual channel, all parties on the path set aside funds in their channels as collateral; to do this they create $O(n^3)$ so-called *virtual* and *bridge* transactions (txs) in total ($O(n^2)$ stored per party, see Sec. 4 for why this is acceptable) that essentially tie the spending of two adjacent base channels into a single atomic action. These txs are kept off-chain. The bridge txs have a *funding* output with the sum of P_1 and P_n's channel balance. To complete channel opening, P_1 and P_n then apply the logic of simple channels (see below) on top of the (off-chain) funding output. LN demands that the funding output is on-chain, but we lift this requirement. We instead guarantee that either endpoint can put the funding output on-chain unilaterally.

In more detail, the channel opening protocol steps are as follows: the two counterparties (i) create new signing keypairs and exchange the resulting public keys, then (ii) if the channel is virtual, they prepare the underlying base channels, next (iii) they exchange signatures for their initial *commitment* txs (see Simple channel below) and lastly, (iv) if the channel is simple, the *funder* (who is the only party that provides coins for a new channel, like in LN) signs and publishes the *funding* tx to the blockchain. A **payment** over an established channel (see Simple channel below) follows a procedure inspired by LN.

Parties $P_1, \ldots, P_n$ can **optimistically close** a virtual channel fully off-chain. At a high level, parties *revoke* their *virtual* txs and the commitment txs, i.e., they cannot use them anymore. This effectively "peels" one *virtualisation layer*. In the process, they redistribute coins so that intermediaries "break even", while P_1 and P_n get their rightful coins (as per the last channel state) in their base channels ((P_1, P_2) and (P_{n-1}, P_n) respectively).

Finally, to **unilaterally close** a virtual channel (P_1, P_n) in case cooperation fails, either P_1 or P_n signs and publishes some txs on-chain. In the simplest case, P_1 publishes her virtual tx. This prompts P_2 to publish her relevant virtual tx and so on up to P_{n-1}. The funding output of (P_1, P_n) is then on-chain and closing can proceed as for simple channels. If instead any intermediary stays inactive, then a timelock expires turning an output into the funding output for (P_1, P_n), at the expense of the inactive party. As discussed later, the funding output employs `ANYPREVOUT` so that the channel needs only a single commitment tx per endpoint, avoiding an exponential state blowup in the recursion depth and making off-chain payments efficient. We assume that parties already have established secure communication channels. In practice, peer discovery can be done as in LN.

In order to build intuition, let us present examples of the lifecycles of a simple and a virtual channel. Consider 5 parties, $A, B, \ldots, E$ and 4 channels, (A, B), $\ldots$, (D, E), that will act as the base of the virtual channel (A, E). We first follow the operations of the simple channel (A, B) and then those of (A, E). We simplify some parts of the protocol to aid comprehension — more details are in [17].

Simple channel. First A and B generate funding and revocation keypairs and exchange public keys (A_F, B_F, A_R, B_R in Fig. 2 – 2 messages). Each then locally generates the *funding* and the two *commitment* txs (F, C_A in Fig. 2). They sign C_A and exchange the signatures (2 messages). A stores her signed commitment tx off-chain and publishes F on-chain. Once F is finalised, the channel is *open*.

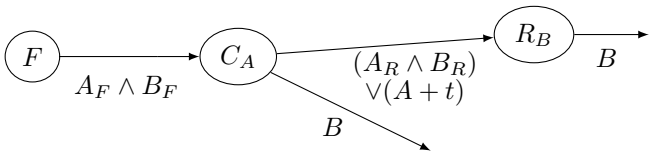


Figure 2: A–B simple channel: Funding (F), A's commitment (C_A) and B's revocation (R_B) txs. $A_F \wedge B_F$ is the funding output. The (symmetric) B's commitment and A's revocation txs are omitted. A_F is A's funding key, A_R her revocation key, $A \wedge B$ needs a signature by both A and B. $A + t$ needs a signature by A after relative timelock t. The top C_A output is spendable either by both A_R and B_R, or by $A + t$, the "or" is denoted with $\vee$.

The funding tx F moves A's initial coins to a 2-*of*-2 *multisig*, i.e., an output that needs signatures from A_F and B_F to be spent. There is one commitment tx per party, stored locally off-chain. The one held by A (C_A, Fig. 2) spends F and has one output for A (initially with all coins) and one for B (initially with 0 coins). A's output can be spent by either a multisig, or by A after a *relative timelock* of t (relative means the countdown starts at the moment of publication). This is so that B has time to *punish* A

if she cheats. B's commitment tx C_B is symmetric. We recommend using $t = 5$ days as a default, the same as LN.

When A *pays* c coins to B, A and B create two new commitment txs. They have the same outputs and scripts as their previous ones, save for the coins: A's outputs have c coins less, B's have c more. They sign them and swap signatures. To ensure only one set of commitment txs is valid, they then revoke their previous ones by building and signing *revocation* txs for the previous commitment txs. B's revocation tx (R_B, Fig. 2) gives B all coins that belonged to A and vice versa. This disincentivises both parties from publishing an old commitment tx under the threat of losing all their coins. A payment needs 3 messages in total. Note that, after generating the new commitment txs and before revoking the old ones, each party has two valid commitments for different channel versions. This is fine, as the payment only concludes when revocation completes. If one party publishes the new commitment tx before revocation, then the other will consider the payment successful and vice versa for the old commitment tx.

To unilaterally *close* (A, B), A or B publish the latest commitment tx on-chain and wait for timelock expiration. The last commitment tx is not revoked, thus punishment is impossible. Simple Elmo channels work like LN simplifed.

Virtual channel. Let channels $(A, B), \ldots, (D, E)$ be open and each "left" party own at least c coins. Let us see how to open the virtual channel (A, E) with A initially having c coins using $(A, B), \ldots, (D, E)$ as base channels. Each base channel may be simple or virtual. If any one is virtual, then (A, E) is recursive. Thanks to the similarity of all layers, the description below is identical in both cases. First, the 5 parties generate and exchange a number of new keys, i.e., all keys that appear in the outputs of Figs. 3–6. Then each base pair removes c coins from the "left" party in their base channel. The updated commitment txs use some of the new keys for the multisig in their input, since, as we will see, so-called *virtual* txs will form a layer, the *virtual layer*, between the funding and the commitment txs from now on (Fig. 7).

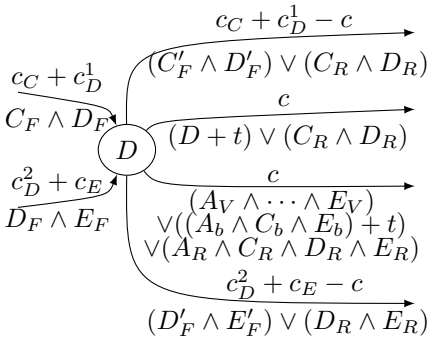


Figure 3: A–E virtual channel: D's initiator virtual tx. Spends the funding outputs of the C–D and D–E channels. D can use it if neither C nor E have published a virtual tx yet. c_D^1/c_D^2: D's coins in C–D/D–E channel. A_V: A's *virtual* key. A_b: A's *bridge* key.

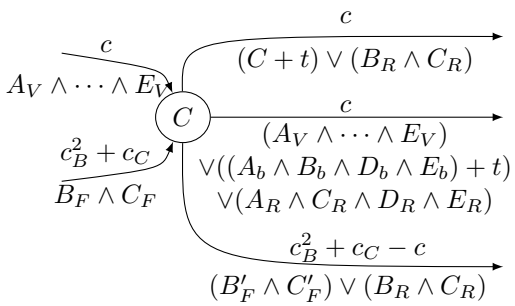


Figure 4: A–E virtual channel: One of C's extend-interval virtual txs. Spends the virtual output of D's initiator tx and the funding output of the B–C channel. C can use it if D has already published its initiator tx and B has not published a virtual tx yet.

Next, each party builds the virtual txs, signs each one with all its keys that appear in the tx's inputs (e.g., B signs the tx of Fig. 4 with B_V and B_F), and exchange the signatures. These txs form Elmo's core. Intuitively, they force each intermediary to interact with both of its base channels instead of one at a time, so that virtual channel information flows along the base channels. There are 3 virtual tx types – looking ahead, we explain them based on how they help close the channel on-chain. Each intermediary (say D) stores a single *initiator* virtual tx, which spends D's two funding outputs and needs to be signed by C_F, D_F, and E_F. It has 4 outputs: one new funding output for each of (C, D) and (D, E) (Fig. 3, 1st & 4th output), one output that refunds the collateral to D (2nd) and, crucially, a so-called *virtual* output (3rd). The latter can be spent by C by the top input of an *extend-interval* tx, the second type of virtual tx (Fig. 4). It needs signatures from the *virtual keys* $A_V \ldots E_V$ of all parties. This tx also spends the other C's funding output, namely that of (B, C) (bottom input). It has 3 outputs: one refunding the collateral to C (top), another virtual output (middle), and a funding output that replaces the one just spent (bottom). Each intermediary stores $n-3$ extend-interval txs, one for each possible interval of parties that have already published their virtual txs[1]. In our example, A also uses its initiator tx (Fig. 5). The format of A's single initiator tx is different. It spends only the funding output of (A, B) and produces 2 outputs: a new funding output for (A, B) (top) and a virtual output (bottom) – E's single initiator tx is symmetric, spending the funding output of (D, E) and producing a new funding output for (D, E) and a virtual output. The two endpoints A, E have only one type of virtual txs, namely the initiator tx.

B can in turn spend A's and C's virtual outputs with a *merge-intervals* tx (Fig. 6), the last virtual tx type, which also needs signatures from $A_V \ldots E_V$. It has 2 outputs: one refunds B and the other produces a new virtual output. The i-th intermediary stores $(i - 2)(n - 1 - i)$ merge-

1. The intermediaries that neighbour endpoints store one $n - 2$ extend-interval txs each.

intervals txs, one for each possible pair of intervals of parties that have already published their virtual txs on the party's left and right[2]. Now all base funding outputs are spent, all intermediaries are refunded and the virtual layer has "collapsed". Additionally, the virtual output of B's tx plays the role of the funding output of the virtual channel (A, E).

The virtual txs have two properties: First, each intermediary can publish at most one virtual tx, which refunds its collateral exactly once. We will see how this is enforced below. Second, if the chain of virtual txs is at any point broken by an inactive intermediary that does not publish its virtual tx, the virtual channel will still be funded correctly and the inactive party will lose its collateral. This is because the unclaimed virtual output turns into the funding output of the virtual channel after a timelock, see, e.g., Fig. 3, output 3, spending method 2.

Keys $A_b \dots E_b$ (Figs. 3–6) can be spent by *bridge* txs, the output of which funds (A, E). Bridge txs allow endpoints to only store a single commitment tx instead of $O(n^3)$. Any virtual output may end up funding the virtual channel, but each virtual output has a different script, thus no single commitment tx is able to spend all of them. Bridge txs fix this by having the same output. This makes virtual channel updates as cheap as simple channel updates.

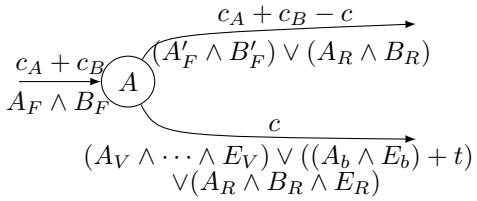


Figure 5: A–E virtual channel: A's initiator virtual tx. Spends the funding output of the A–B channel. A can use it if B has not published a virtual tx.

Some considerations remain to ascertain the scheme's security. Firstly, if all virtual outputs were identical, an adversary could fabricate arbitrarily many virtual outputs using its own, external to the protocol, coins and publish a perpetual stream of merge-intervals txs that spend one valid and one fabricated virtual output. This would prevent an honest endpoint from accessing their funding output. We avoid this by specifying on each virtual output the exact sequence of parties that have already published a virtual tx and only allowing the parties at the two edges of the sequence to extend it with their virtual tx. If all intermediaries publish a virtual tx, then the last published one cannot be spent by another virtual tx. This ensures that the endpoints will eventually obtain a funding output, but needs intermediaries to store $O(n^2)$ virtual txs for a virtual channel over n parties. Secondly, we have to carefully select the values of timelocks to ensure that each party has enough time to act. Timelocks increase linearly with the recursion depth. The exact values are shown in [17].

2. The intermediaries that neighbour endpoints store $n - 3$ merge-intervals txs each.

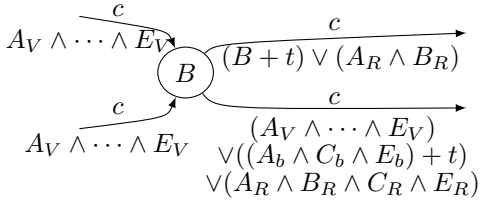


Figure 6: A–E virtual channel: One of B's merge-intervals virtual txs. Spends the virtual outputs of A's and C's virtual txs. B can use it if both A and C have already published their initiator or extend-interval tx.

We now return to the opening procedure. After the 5 parties set up the virtual txs, they revoke their previous commitment txs. They do this by signing the relevant revocation txs, just like for a simple channel. It takes $12 \cdot (n - 1)$ messages, i.e., 6 messages per endpoint and 12 messages per intermediary, to set up the virtual layer.

At last, the virtual layer is set up: Both A and E can unilaterally force a funding output for channel (A, E) on-chain, no matter what the rest of the parties do. Likewise, honest intermediaries can unilaterally retrieve their funds.

A and E finally exchange commitment transactions for their new channel, thus concluding its opening. A and E can pay each other over their virtual channel exactly like they would over a simple channel; we refer the reader to the relevant description above.

Today, every signature of a Bitcoin tx T must sign the hash of all txs spent by T (more precisely, the spent *outpoints*), thus we currently cannot store a signed tx T off-chain and decide only later which on-chain tx(s) it will spend. An `ANYPREVOUT` signature fixes this by not including this hash in the signed message. This enables a tx signed with an `ANYPREVOUT` signature to spend any output as long as it has a suitable script, even an output that was unknown at the time of signing.

Funding outputs use the `ANYPREVOUT` flag, thus ensuring that a single pair of commitment txs can spend any of the funding outputs. If `ANYPREVOUT` were not used, each virtual layer would need a copy of the entire set of discussed txs for each possible funding output of its base layer, resulting in exponential storage requirements. To make matters worse, a payment over (A, E) would need A and E to renegotiate exponentially many commitment txs, as well as recalculate all their downstream txs, which would in turn need interaction with intermediaries of all virtual channels built over (A, E), completely defeating the essence of payment channels.

Cooperative closing. To enhance usability, our protocol enables closing the virtual channel off-chain if all parties cooperate. To do this, the endpoints first let the intermediaries know their final virtual channel balance. Then the parties of each base channel create new commitment txs for their channels, moving the collateral back into the channel: the

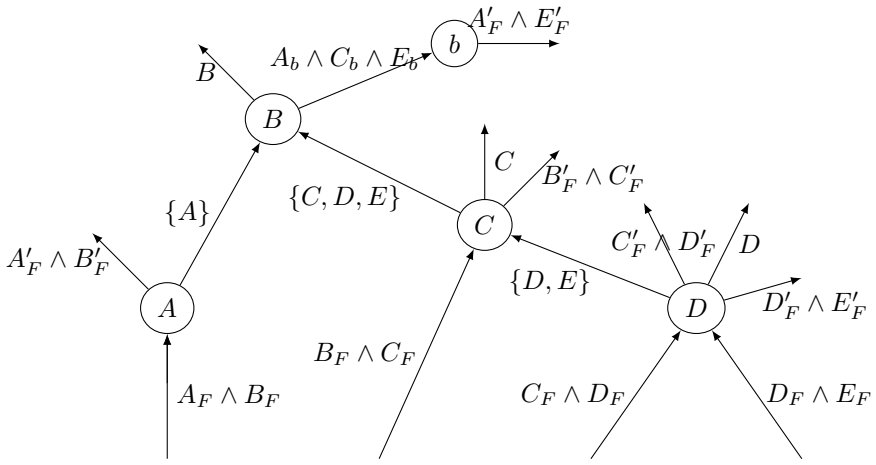


Figure 7: Example of on-chain (unilateral) closing of virtual channel (A, E), built on top of simple channels $(A, B), \ldots, (D, E)$. A and D start closing by publishing their initiator virtual txs (Fig. 5 and 3 respectively), then C publishes its suitable extend-interval virtual tx (Fig. 4), after which B publishes its suitable merge-intervals virtual tx (Fig. 6). No party stays inactive. The virtual transactions A–D form the virtual layer. Virtual outputs are marked with the set (interval) of parties that have already published a tx. ***Bridge*** txs such as b are used by A and E to convert the various virtual outputs into the same funding output $A'_F \wedge E'_F$, as `ANYPREVOUT` only works across identical outputs.

"left" party gets A's coins and the "right" one gets E's. Thus all intermediaries "break even" across their two channels. Once they do this, the parties revoke all virtual txs, using a logic similar to the revocation procedure of simple channels but scaled up to all parties. This is why all virtual tx outputs (Figs. 3–6) have a spending method with $A_R \ldots E_R$ keys.

What if one party does not cooperate? Then one or more of the others must close unilaterally on-chain. Fig. 7 shows how this would play out if A and D initiate this procedure.

Our protocol is recursive because both simple and virtual channels are ultimately represented by a funding output that either already is or can be put on-chain, thus new virtual channels can be built on either. Our channels avoid key reuse on-chain, thus ensuring party privacy from on-chain observers. More details are in [17].

Remark. We conjecture that without `ANYPREVOUT` no efficient virtual channel over Bitcoin can be built. In particular, if any protocol (i) has an efficient closing operation with $O(1)$ on-chain transactions, (ii) has parties locally store the channel state as transactions and signatures and (iii) does not need locking up coins as collateral (as [2] does), then each party needs space exponential in the number of intermediaries. The second requirement is natural, since, to our knowledge, all trustless layer 2 protocols over Bitcoin require all implicated protocol parties to actively sign off every state transition and locally store their counterparties' relevant signatures to ensure they can unilaterally exit later.

3. Security

Before providing the UC-based security guarantees, it is useful to obtain concrete properties directly from our protocol, essentially providing standard property-based security in addition to our UC guarantees. We first delineate the security guarantees Elmo provides by proving Lemma 1 which discusses the conservation of funds. The formal statements along with all proofs are deferred to [17]. Informally, it establishes that if an honest was implicated in a channel that has now been unilaterally closed, then the party will have at least the expected funds on-chain.

Lemma 1 (Real world balance security (informal)). Consider a real world execution with $P \in \{Alice, Bob\}$ an honest, non-negligent party. Assume that all of the following are true:

- P opened the channel, with initial balance c,
- P is the funder of n channels, each funded with f_i coins,
- P has cooperatively closed k channels, where the i-th channel transferred r_i coins from the funded virtual channel to P,
- P has sent m payments, each involving d_i coins,
- P has received l payments, each involving e_i coins.

If P performs a successful unilateral close, eventually there will be $h \in \mathbb{N}^*$ outputs on-chain spendable only by P, each of value c_i, such that

$$\sum_{i=1}^{h} c_i \geq c - \sum_{i=1}^{n} f_i + \sum_{i=1}^{k} r_i - \sum_{i=1}^{m} d_i + \sum_{i=1}^{l} e_i \ . \quad (1)$$

The expected funds are [initial balance - funds for funded virtuals + funds returned from funded virtuals - outbound payments + inbound payments]. A similar guarantee is provided for the ideal-world functionality in [17].

Proof Sketch [Lemma 1]: All execution paths are followed, keeping track of the resulting balance in each case and concluding that balance is secure in all cases, except if signatures are forged. □

Importantly, our protocol Π_{Chan} in fact provides a stronger guarantee: a party can always successfully unilaterally close its channel and obtain the expected funds on-chain within a known number of blocks. This guarantee is sufficient to make Elmo reliable enough for real-world applications. However an ideal world functionality with such guarantees would have to track specific txs and signatures, making it as complicated as the protocol, thus violating the spirit of the simulation-based security paradigm.

$\mathcal{G}_{\mathrm{Chan}}$ [17] halts on security breaches (e.g., lower than expected balance). We prove the "no-halt" Lemma which informally states that if an ideal party is honest, $\mathcal{G}_{\mathrm{Chan}}$ does not halt with overwhelming probability.

4. Efficiency Evaluation & Simulations

We offer a cost and efficiency comparison of this work with LVPC [16] and Donner [2]. We focus on these due to their exclusive support of virtual channels over any number of base channels. LVPC achieves this via recursion and Donner via its variadic property (cf. Table 2).

We count the communication, storage and on-chain cost of a virtual channel in each protocol. We also simulate the execution of a large number of payments among many parties and derive payment latency and fees – We thus obtain an end-to-end understanding of both the requirements and benefits of each protocol.

Cost calculation. Consider the setting of 1 funder (P_1), 1 fundee (P_n) and $n-2$ intermediaries ($P_2, \ldots, P_{n-1}$) where P_i has a base channel with each of P_{i-1}, P_{i+1}. We compare the costs of off-chain opening (Table 3) and on-chain closing unilaterally (Table 4). Comparison details and choices are deferred to [17].

Regarding opening, in Table 3 we calculate for each of the 3 protocols the number of communication rounds required, the total size of outgoing messages as well as the amount of space for storing channel data. We calculate funder, fundee and intermediary requirements, along with the aggregate for all parties.

Regarding closing, in Table 4 we calculate for each of the three protocols the worst-case on-chain cost for a party in order to unilaterally close its channel. The cost is measured both in the number of transactions and in their total size.

For each endpoint (funder and fundee), we show the worst-case cost of unilaterally closing the virtual channel (i.e., one initiator virtual tx, one bridge tx, and one commitment tx), whereas for each intermediary we report the worst-case cost of closing a base channel (i.e., one merge-interval virtual tx). We also present the worst-case total on-chain cost aggregated over all parties. Note that the latter cost is not simply the sum of the worst-case costs of all parties, as one party's worst case is not necessarily the worst case of another. This cost rather represents the maximum possible load an instantiation of each protocol could add to the blockchain when closing.

We note that Elmo exploits MuSig2 [18], [21] to reduce both its on-chain and storage footprint: the n signatures that are needed to spend each virtual and bridge output can be securely reduced to a single aggregate signature. The same cannot be said for Donner, since this technique cannot optimise away the n outputs of the funder's transaction $\mathtt{tx}^{\mathtt{vc}}$. Likewise LVPC cannot gain a linear improvement with this optimisation, since each of its relevant transactions ("split", "merge" and "refund") needs constant signatures.

We furthermore note that, since human connections form a small world [19], we expect that in practice the need for virtual channels with a large number of intermediaries will be exceedingly rare. This is corroborated by the fact that LN only supports payments of up to 20 hops without impact to its usefulness. Therefore, the asymptotic network and storage complexity are not as relevant as the concrete costs for specific, low values of n. Under this light, the overhead of Elmo is tolerable. For example, the total size of messages sent and received for a funder when opening an Elmo channel of length $n = 6$ are less than 3 times those of Donner and slightly cheaper than LVPC (Table 3). Moreover, Elmo can be deployed on commodity hardware: To set up and keep a channel of length $n = 6$ active, an intermediary needs to send less than 14 KiB over the network and store less than 15 KiB — the endpoints need even less resources.

Payment simulations. We implemented a simulation framework[3] in which a list of randomly generated payments are carried out. A simulation run is parametrised by a list of payments (sender, receiver, value triples), the protocol (Elmo, Donner, LVPC, LN or on-chain only), which future payments each payer knows and the utility function it maximises. Our framework is of independent interest, as it can be used for evaluating various payment networks.

We here show the performance of the 3 protocols with respect to the metrics payment channels aim to improve, namely payment latency and fees (Table 5). We have designed 3 workloads: "power law", in which incoming payments follow a zipf [24] distribution, "preferred receiver", in which each party has a preferred payee which receives half of the payments, and "uniform", where payments are chosen uniformly at random. More details are deferred to [17].

As evidenced, Elmo is the best or on par with the best protocol in every case. We attribute this to the flexibility of Elmo, as it is both variadic and recursive, thus able to exploit the cheapest payment method in all scenarios. In particular, Donner is consistently the most fee-heavy protocol and LVPC the slowest. Elmo experiences similar delays to Donner and slightly higher fees than LVPC.

5. Related Work

Perun [11], Thunderdome [3], and GSCN [12] build virtual state channels on the Turing-complete language of Ethereum, which cannot be translated to Bitcoin due to its lower expressiveness. Similar features are provided by Celer [10]. Hydra [6] provides state channels for Cardano [7].

3. https://gitlab.com/orfeas-litos/virtual-channels-simulation

Table 2: Features & requirements comparison of virtual channel protocols

	Unlimited lifetime	Recursive	Variadic	Symmetric	Script requirements
LVPC [16]	✗	◐[a]	✗	✓	Bitcoin
BCVC [1]	✓	✗	✗	✓	Bitcoin
Perun [11]	✓	✗	✗	✓	Ethereum
GSCN [12]	✓	✓	✗	✓	Ethereum
Donner [2]	✗	✗	✓	✗	Bitcoin
this work	✓	✓	✓	✓	Bitcoin + ANYPREVOUT

a. lacks security analysis

	Open										
	Funder			Fundee			Intermediary			Total	
	party rounds	size		party rounds	size		party rounds	size		size	
		sent	stored		sent	stored		sent	stored	sent	stored
LVPC	$8(n-2)$	$1381(n-2)$	$3005(n-2)$	7	1254	2936	16	2989	6385	$4370n-8740$	$9390n-18780$
Donner	2	$184n+829$	$1332.5k+$ $43n+125.5$	1	$43n+192.5$	$1332.5k+$ $43n+125.5$	1	547	$1332.5k+$ $43n+125.5$	$774n-71$	$1332.5kn+$ $43n^2+125.5n$
Elmo	6	$32n^3-128n^2$ $+544n-276$	$\frac{128}{3}n^3-128n^2$ $+\frac{1276}{3}n+220$	6	$32n^3-128n^2$ $+544n-340$	$\frac{128}{3}n^3-128n^2$ $+\frac{1276}{3}n+220$	12	$96n^3-256n^2$ $+404n-40$	$96n^3-256n^2$ $+468n+88$	$96n^4-384n^3+$ $724n^2+240n-792$	$96n^4-\frac{1088}{3}n^3+$ $660n^2+\frac{8}{3}n+520$

Table 3: Open efficiency comparison of virtual channel protocols with n parties and k payments.

	Unilateral Close							
	Intermediary		Funder		Fundee		Total	
	#txs	size	#txs	size	#txs	size	#txs	size
LVPC	3	627	2	383	2	359	$2n-2$	$435n-510.5$
Donner	1	204.5	4	$704+43n$	1	136.5	$2n$	$458n-26$
Elmo	1	297.5	3	376	3	376	$n+1$	$254.5n-133$

Table 4: On-chain worst-case closing efficiency comparison of virtual channel protocols with n parties.

	Delays (sec) and Relative Fees					
	Power Law		Preferred Receiver		Uniform	
	Delay (sec)	Fees	Delay (sec)	Fees	Delay (sec)	Fees
LVPC	2873	0.81	2440	0.80	3589	0.99
Donner	2842	1	2361	1	3589	1
Elmo	2864	0.83	2344	.87	3589	0.99

Table 5: Average per-payment delay (both on- and off-chain) and relative fee. Less is better. Blue is best, red is worst.

Several works propose virtual channel constructions for Bitcoin. LVPC [16] enables a virtual channel to be opened on top of two preexisting channels, but only provides fixed duration channel lifetime. Bitcoin-Compatible Virtual Channels [1] enables virtual channels on top of two preexisting simple channels and offers two protocols, the first of which guarantees that the channel will stay off-chain for an agreed period, while the second allows the single intermediary to turn the virtual into a simple channel. This strategy has the shortcoming that even if it is made recursive (a direction left open in [1]) after k applications of the constructor the virtual channel participant will have to publish on-chain k transactions in order to close the channel if all intermediaries actively monitor the blockchain.

Donner [2] (released originally concurrently with the first technical report of Elmo) achieves variadic, but not recursive virtual channels. It is more efficient than Elmo in terms of storage, computation and communication, and boasts a simpler design. On the downside, Donner has the funder lock as collateral twice the amount of the useful channel funds: once on-chain with funds external to the base channels and once off-chain within its base channel. The on-chain coins are unusable, creating opportunity cost. The funder's collateral in Donner is thus double that of LVPC or Elmo. The collateral for all other parties is the same across LVPC, Donner, and Elmo. Additionally, contrary to Elmo, a Donner channel needs active periodic collaboration of all channel parties to refresh it, thus Donner channels do not have an unlimited lifetime. We conjecture that using external coins precludes variadic virtual channels with unlimited lifetime. This further means that Donner is not symmetric. Donner also uses placeholder outputs which, due to the minimum coins they need to exceed Bitcoin's *dust limit*, may skew the incentives of rational players and add to the channel cost. Its lack of recursiveness means coins in a Donner channel can be paid to someone beyond the counterparty only if all base parties collaborate or by going on-chain. Further, its design complicates future iterations that lift its current restriction that only one of the two channel parties can fund the virtual channel. Their work also introduces *Domino*, which we discuss in [17]. We briefly mention here that, contrary to what is claimed in [2], Domino does not force the base channels of an Elmo virtual channel to close.

We refer the reader to [17] for further related work, including a technical issue in Donner and its resolution. Table 2 contains a comparison of the features and limitations of virtual channel protocols, including the current work. Future work can be found in [17].

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